

SynthMicro — Whole Blood Smear XAI Report

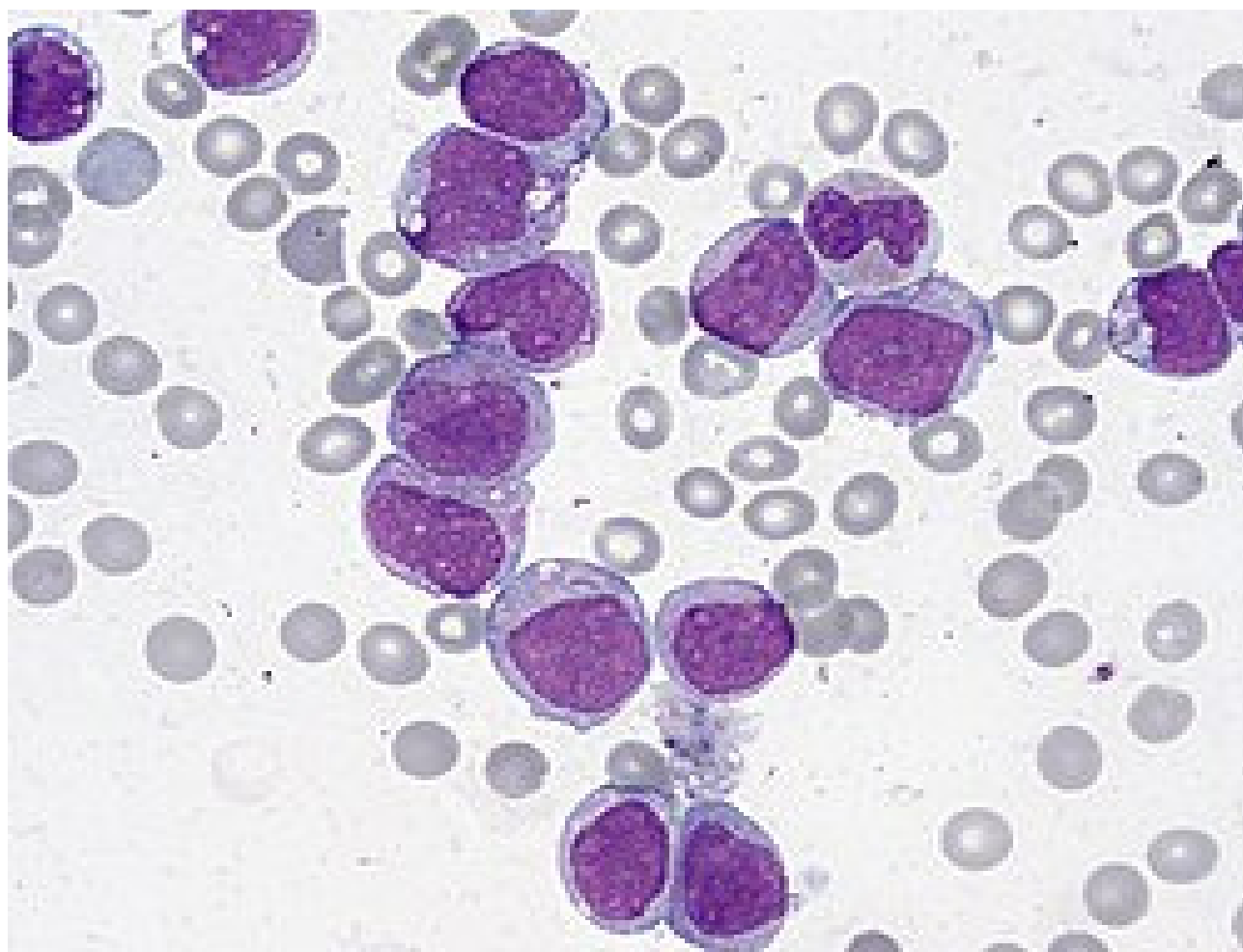
Analysis ID: 8fb9e05f714e42659b12590c882643f3
Generated: 2026-09-07 19:51:08

Pipeline: Cellpose segmentation → 384×384 cell extraction → ResNet50 classification → Grad-CAM → SHAP → explanatory report.

Measurement	Result
Cellpose objects	83
Nucleated-cell candidates	16
Myeloblast-like candidates	9
Myeloblast-like proportion	56.25%

Screening interpretation: the proportion above is a research-oriented model indicator based on candidate cells selected by the notebook's purple-score rule. It is not a cancer probability and does not establish AML or another diagnosis.

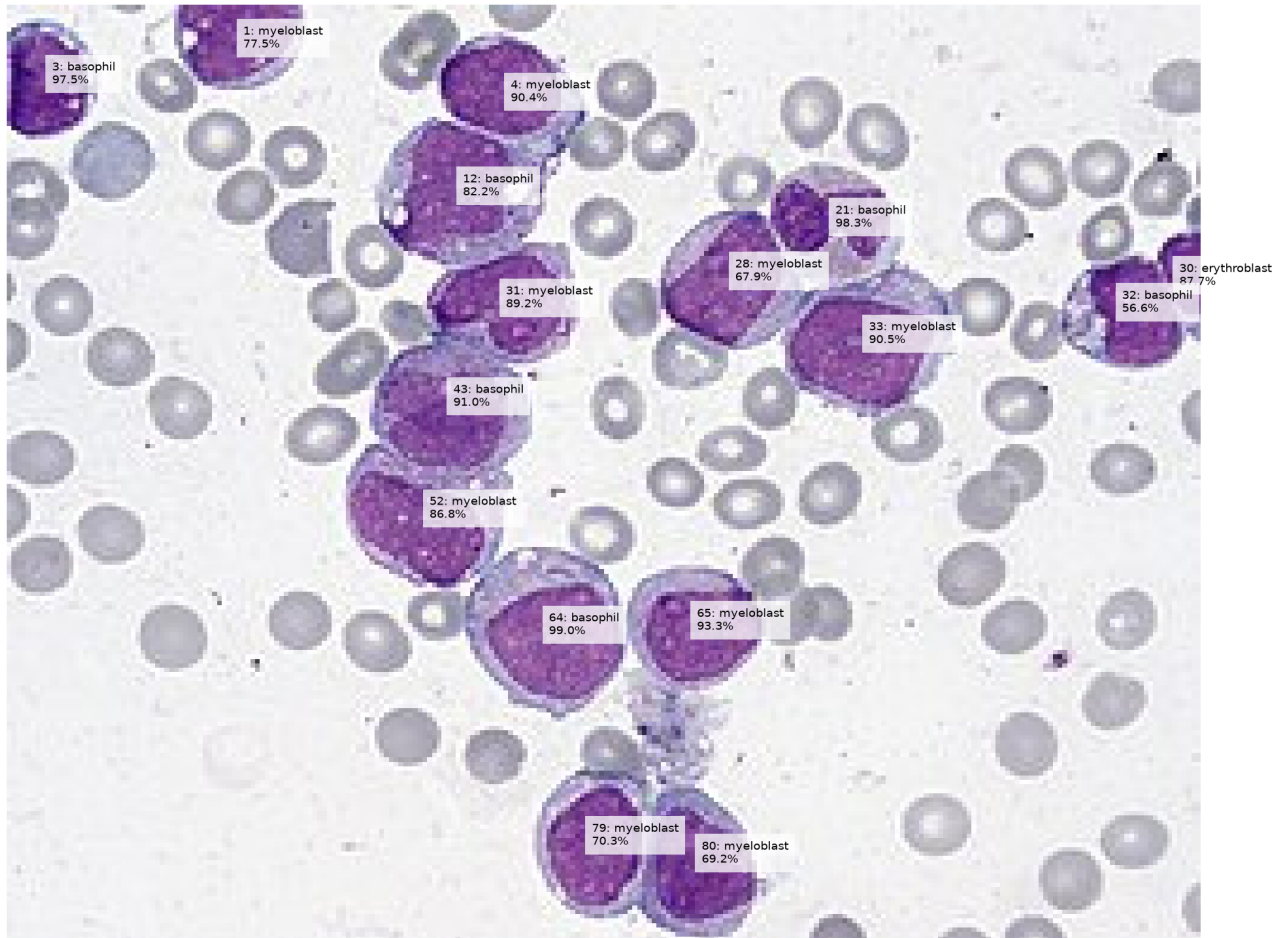
Whole Smear



Original uploaded blood-smear image analyzed by Cellpose.

Segmentation and Candidate Overlay

Whole Smear — Candidate Cell Classification



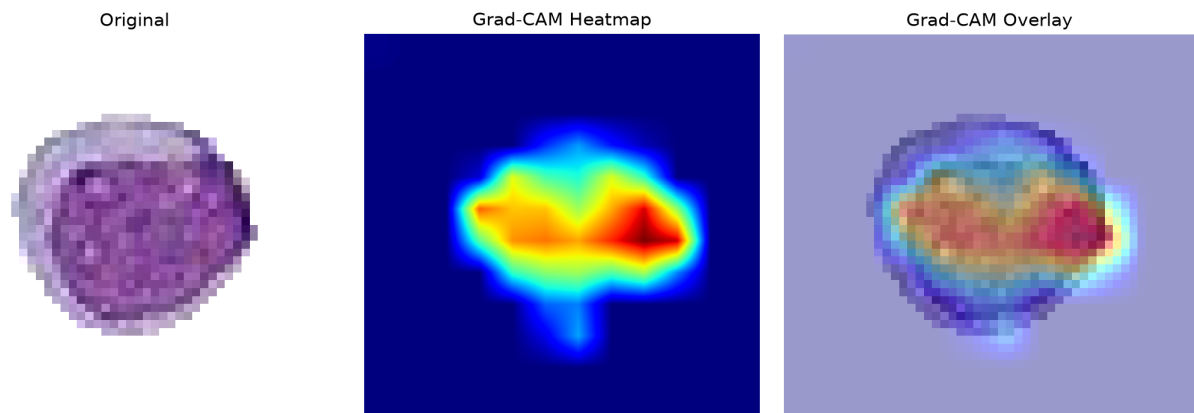
Cell-Level Results

ID	Area	Purple	Class	Confidence
1	446	53.5	myeloblast	77.5%
3	401	57.1	basophil	97.5%
4	598	52.0	myeloblast	90.4%
12	899	47.0	basophil	82.2%
21	606	42.4	basophil	98.3%
28	698	49.1	myeloblast	67.9%
30	149	60.2	erythroblast	87.7%
31	647	52.8	myeloblast	89.2%
32	501	51.4	basophil	56.6%
33	895	47.4	myeloblast	90.5%
43	795	49.2	basophil	91.0%
52	821	48.8	myeloblast	86.8%
64	989	43.8	basophil	99.0%
65	631	50.2	myeloblast	93.3%
79	659	49.2	myeloblast	70.3%
80	688	51.1	myeloblast	69.2%

Grad-CAM Explanations

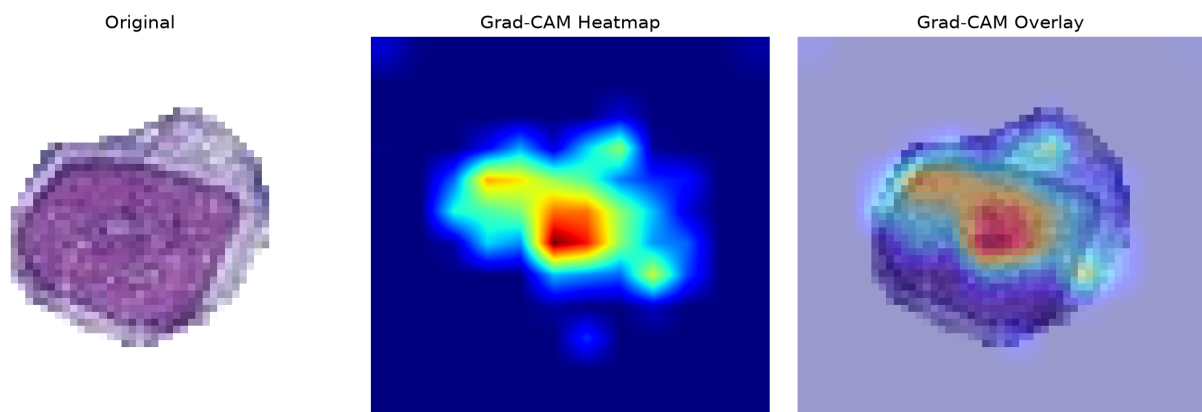
Grad-CAM shows regions contributing to the selected model class. It is an explanation of model behavior, not a direct measurement of a biological structure.

Cell 65 — myeloblast (93.3%)



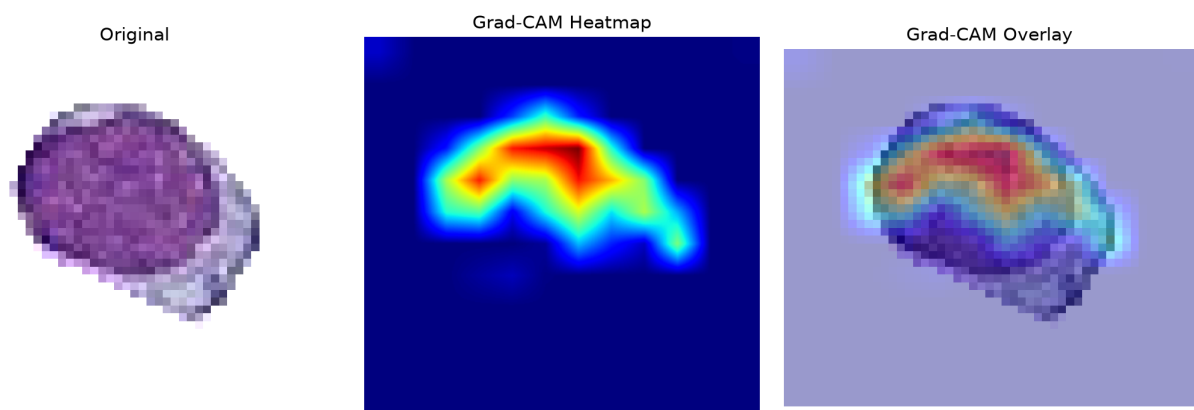
The ResNet50 model classified this segmented candidate as a myeloblast with 93.3% confidence. This is a model-level cell classification; a myeloblast-like prediction alone does not establish AML or another cancer diagnosis.

Cell 33 — myeloblast (90.5%)



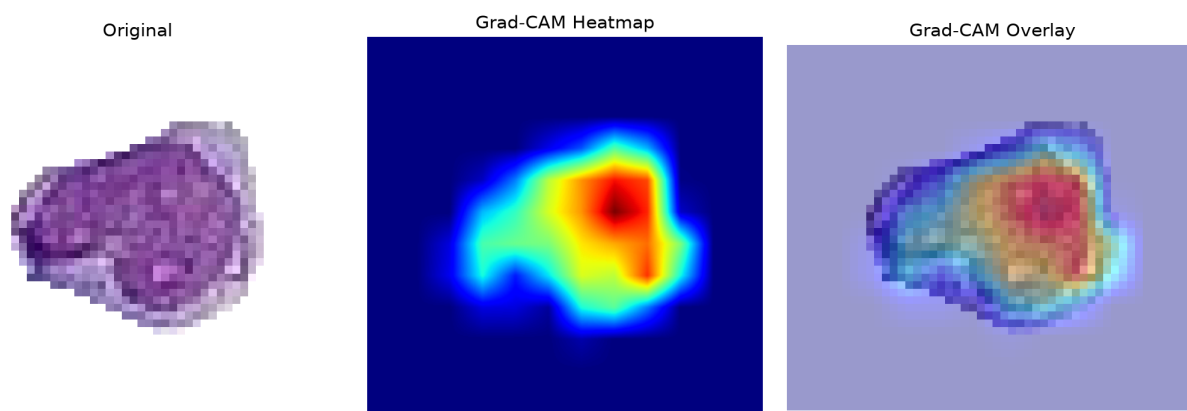
The ResNet50 model classified this segmented candidate as a myeloblast with 90.5% confidence. This is a model-level cell classification; a myeloblast-like prediction alone does not establish AML or another cancer diagnosis.

Cell 4 — myeloblast (90.4%)



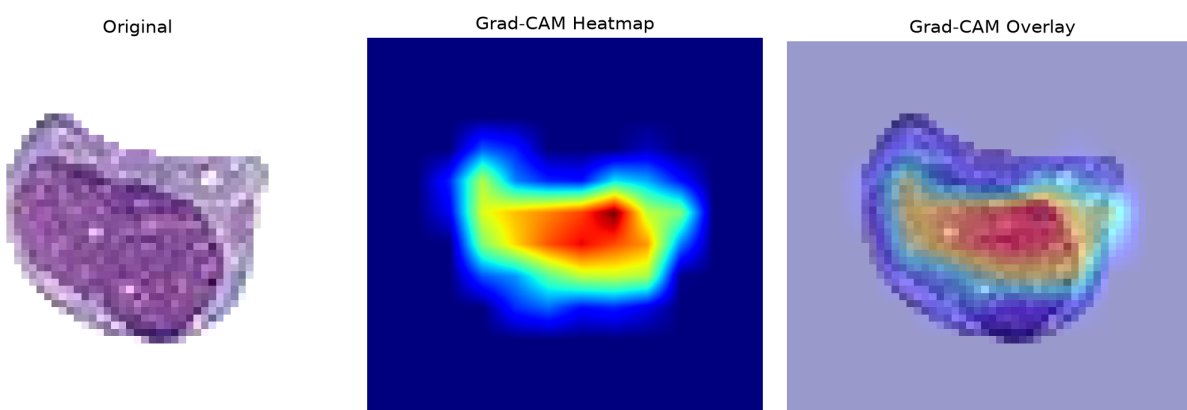
The ResNet50 model classified this segmented candidate as a myeloblast with 90.4% confidence. This is a model-level cell classification; a myeloblast-like prediction alone does not establish AML or another cancer diagnosis.

Cell 31 — myeloblast (89.2%)



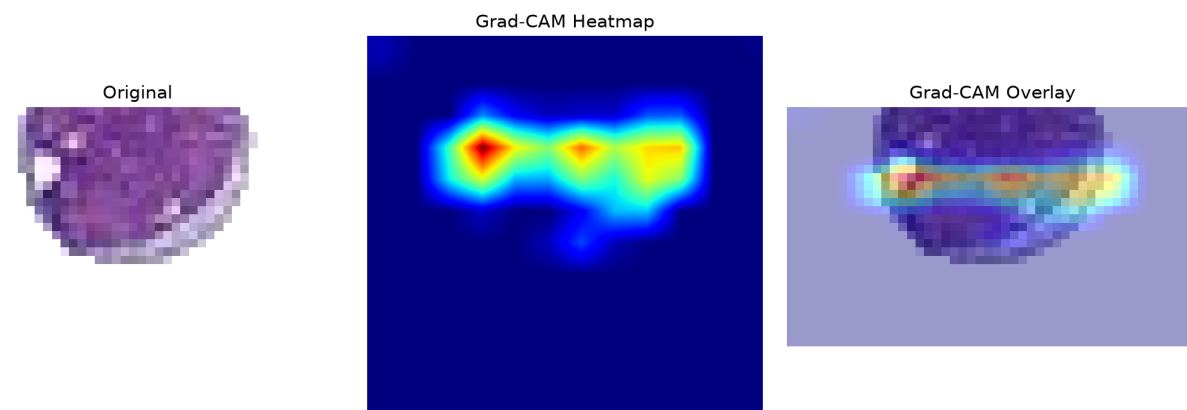
The ResNet50 model classified this segmented candidate as a myeloblast with 89.2% confidence. This is a model-level cell classification; a myeloblast-like prediction alone does not establish AML or another cancer diagnosis.

Cell 52 — myeloblast (86.8%)



The ResNet50 model classified this segmented candidate as a myeloblast with 86.8% confidence. This is a model-level cell classification; a myeloblast-like prediction alone does not establish AML or another cancer diagnosis.

Cell 1 — myeloblast (77.5%)



The ResNet50 model classified this segmented candidate as a myeloblast with 77.5% confidence. This is a model-level cell classification; a myeloblast-like prediction alone does not establish AML or another cancer diagnosis.

SHAP Explanations

SHAP attribution magnitude summarizes the magnitude of pixel-level contribution toward the selected class. It is a model explanation, not a clinical measurement.

SHAP output was unavailable for this analysis.

Interpretation and Limitations

- The classifier has five closed-set classes: basophil, erythroblast, monocyte, myeloblast and segmented neutrophil.
- RBCs and unknown cell types are not explicit classes and can therefore be forced into one of the five outputs.
- Cellpose segmentation quality affects every downstream result.
- Stain, microscope, illumination and acquisition differences can cause domain shift.
- Confidence is not a calibrated probability of disease.
- A myeloblast-like prediction is not equivalent to an AML diagnosis.
- Clinical assessment requires appropriate hematopathology and laboratory testing.